

Janadri Yalla Yashwanth

Machine Learning Engineer | MLOps · Model Training · Docker · CI/CD

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SUMMARY

Machine learning engineer and 2025 CS (AI & ML) graduate who trains deep-learning models and ships them to production. I train custom object detectors (YOLO) on self-labeled data and deploy production detection models (SAM-2, DeepForest, SegFormer) behind FastAPI, owning the path from model to Docker image, GitHub Actions CI/CD, and systemd GPU/CPU workers. Comfortable across the ML lifecycle — data pipelines, training, validation, containerization, and deployment. B.Tech in CS (AI & ML), CGPA 8.6; immediate joiner.

TECHNICAL SKILLS

Languages: Python, SQL, C, Java

ML & Deep Learning: PyTorch, TensorFlow, scikit-learn, YOLO (Ultralytics), SAM-2, DeepForest, SegFormer, OpenCV

ML Lifecycle / MLOps: Model training & validation, custom dataset labeling & augmentation, data pipelines, model packaging & serving

Deployment & CI/CD: Docker, Docker Compose, GitHub Actions, CI/CD, systemd, Linux

Cloud & Data: AWS S3, GDAL / rasterio, PostgreSQL, Redis, Celery

Algorithms: Classification, regression, clustering, object detection, deep learning, NMS

Practices: Git version control, testing & debugging, Agile collaboration

Tools: Git & GitHub, Claude Code, VS Code, Jupyter, Hugging Face Hub, Docker, GitHub Actions, Linux

EXPERIENCE

Software Engineer Trainee — Marut Drones

Feb 2026 – Present

- Productionized three ML detection pipelines (SAM-2, DeepForest, GroundingDINO/LangSAM) behind a FastAPI backend with async Celery/Redis job orchestration, taking models from prototype to scalable serving.
- Owned the ML deployment path end to end — packaged the PyTorch model stack into reproducible Docker images and shipped them through GitHub Actions CI/CD to systemd-managed GPU and CPU workers.
- Engineered data pipelines that stream cloud-optimized rasters from S3 via GDAL /vsi3/ and rasterio, with elevation-based post-processing and cross-tile NMS deduplication for reliable model outputs.
- Debugged production model failures (CUDA/PyTorch mismatches, GPU-versus-CPU serving, failed migrations) to keep inference stable and reduce cycle time.

Machine Learning Intern — Rooman Technologies

Jan 2025 – Mar 2025

- Trained and evaluated a sleep-disorder classification model (TensorFlow) on physiological and behavioral data, reaching 98% accuracy, and delivered it through a web app.

SELECTED PROJECTS

Custom Object Detection with YOLO

Python, YOLO (Ultralytics), PyTorch, OpenCV, custom datasets

- Trained and validated custom YOLO detectors on self-labeled datasets for number-plate, vehicle, water-body, and broken-fence detection.
- Owned the full training pipeline — data collection, annotation, augmentation, train/validation splits, and per-task tuning.

Aerial Image Object Detection — Multi-Model Benchmark

Python, PyTorch, Hugging Face Transformers, SegFormer, OpenCV

- Benchmarked 4+ segmentation models on identical drone imagery and selected SegFormer-B5 (ADE20K) for the cleanest detections, cutting false positives roughly 3x versus baselines.
- Built the OpenCV/NumPy inference pipeline — tiling, filtering, mask-to-instance extraction, and per-class NMS — with annotated image and JSON outputs.

EDUCATION

Sri Venkateswara College of Engineering, Tirupati

Dec 2021 – May 2025

B.Tech in Computer Science (Artificial Intelligence & Machine Learning) — CGPA: 8.6

Narayana Junior College, Hyderabad

Jun 2019 – Mar 2021

Intermediate (MPC) — 96%

CERTIFICATIONS

- [CS50's Introduction to Programming with Python — Harvard University Online](#)
- [Databases: Topics in SQL — Stanford Online](#)